Integrated microwave Doppler system with FPGA and wireless communication system for tracking near-Earth objects

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Introduction

Near-Earth Objects (NEOs), such as asteroids and meteoroids, pose a potential threat to Earth due to their unpredictable trajectories and high velocities. Early detection of these objects is therefore crucial for scientific observation, risk assessment, and planetary defense. This project presents the design and implementation of an integrated space-to-ground monitoring system that utilizes Doppler-based microwave sensing to detect and characterize the approach of a meteoroid toward Earth.

The system operates by deploying a microwave Doppler sensor in space capable of emitting electromagnetic waves toward the surrounding environment. When a meteoroid passes through the sensor's field of detection, the reflected waves undergo a frequency shift proportional to the object's relative velocity. This Doppler shift is captured and converted into an electrical signal, which is then processed by an onboard Field-Programmable Gate Array (FPGA). The FPGA digitizes the raw sensor data, applies signal-processing algorithms to extract velocity information, and encrypts the resulting data for secure transmission.

The encrypted data is then transmitted from the space module to a ground-based receiving station using an RF communication link. On Earth, an RF receiver captures the incoming signal and transfers it to a computer system that performs decryption, signal reconstruction, and analysis. The processed information can then be used to determine the meteoroid's speed, trajectory, and potential risk.

By integrating sensing, signal processing, encryption, and wireless communication into a single system, this project demonstrates a modern, efficient approach to real-time detection of celestial objects. It further provides a framework suitable for deployment on small satellite platforms, contributing to advancements in space situational awareness and planetary protection technologies.

Objectives

The primary objective of this project is to develop an integrated space-to-ground monitoring system capable of detecting and analyzing the approach of a meteoroid using Doppler-based microwave sensing and secure wireless communication. To achieve this overarching goal, the project aims to fulfill the following specific objectives:

1. **Detect Meteoroid Motion Using Doppler Effect**
   Implement a microwave Doppler sensor in space to measure frequency shifts resulting from the relative motion of a meteoroid.
2. **Process and Analyze Sensor Data Using FPGA Technology**
   Utilize an FPGA to digitize incoming signals, perform Doppler frequency analysis, and extract key parameters such as velocity and direction.
3. **Implement Secure Data Encryption**
   Develop and integrate a robust encryption mechanism within the FPGA to ensure the confidentiality and integrity of transmitted scientific data.
4. **Transmit Data Reliably via RF Communication**
   Establish a wireless RF link capable of sending encrypted data from the space module to a ground receiver with minimal loss and high reliability.
5. **Receive and Decode Signals at the Ground Station**
   Equip the Earth-based station with an RF receiver and PC system to decrypt, reconstruct, and interpret the transmitted information.
6. **Provide Accurate Real-Time Assessment of Meteoroid Proximity**
   Deliver real-time results that allow estimation of meteoroid velocity, trajectory, and potential threat level.
7. **Demonstrate a Compact and Energy-Efficient Solution**
   Ensure that the system is optimized for low power consumption and suitable for deployment on small satellite platforms or CubeSat missions.

Theory

The operation of this project is grounded in several scientific and engineering principles that enable the detection of meteoroids and the reliable transfer of information from space to Earth. The first foundational concept is the Doppler effect, which describes how the frequency of electromagnetic waves changes when they are reflected from a moving object. When the microwave signal emitted by the sensor encounters a meteoroid, its motion causes a shift in the frequency of the returned signal. This shift carries information about the object's velocity and direction relative to the sensor. By analyzing this frequency variation, the system can determine whether a meteoroid is approaching and estimate its movement characteristics.

Another essential component of the theory is the use of microwave Doppler sensing technology. Microwave signals are highly suitable for space applications because they propagate efficiently, penetrate dust and small debris, and maintain stability in harsh environmental conditions. The Doppler sensor converts the reflected waves into an electrical signal that represents the motion of the detected object. This electrical signal is continuous and analog in nature, meaning it must be processed and interpreted digitally to extract useful information.

The FPGA (Field-Programmable Gate Array) serves as the primary processing unit within the system, and its theoretical foundation lies in digital signal processing. FPGAs allow parallel computation, making them capable of handling high-speed, real-time data streams from the Doppler sensor. Inside the FPGA, the incoming signal undergoes filtering, frequency analysis, and data preparation. The FPGA then encrypts the processed data, ensuring that the information sent from space to Earth remains secure. Hardware-based encryption is particularly effective in this environment because it provides strong security with minimal delay and power consumption.

Radio frequency communication represents another theoretical pillar of the system. RF links are commonly used in space communications because they can transmit data over long distances without relying on physical connections. The RF transmitter in the space module modulates the digital, encrypted data onto a carrier wave and sends it toward Earth. The signal travels through free space, gradually weakening due to distance, but it remains detectable by sensitive RF receivers on the ground.

At the ground station, the RF receiver plays a crucial role by capturing the transmitted signal and converting it back into a usable digital format. This

demodulated data is then passed to a computer system, where the theory of digital communication continues with decryption and analysis. The PC executes the reverse of the onboard encryption process to recover the original processed information from the FPGA. Once the data is decrypted, it can be interpreted to determine the meteoroid's proximity and movement, completing the entire sensing-to-analysis process.

Together, these theoretical elements form a complete chain of scientific principles that enable the system to detect a moving object in space, process its signal intelligently, secure the data, transmit it across long distances, and reconstruct the information accurately on Earth.

Equipment

The successful implementation of this project requires a set of specialized electronic, communication, and computational components. Each piece of equipment plays a critical role in enabling the detection, processing, transmission, and reception of meteoroid-related data.

1. Microwave Doppler Sensor

A continuous-wave microwave motion sensor capable of emitting and receiving electromagnetic waves. It detects frequency shifts caused by moving objects such as meteoroids and outputs an analog electrical signal representing the Doppler response.

2. Field-Programmable Gate Array (FPGA) Board

A reprogrammable digital hardware platform used to perform high-speed signal processing, data encryption, and communication control. The FPGA handles real-time frequency analysis and prepares the data for transmission.

3. Analog-to-Digital Converter (ADC)

A module integrated within or connected to the FPGA to convert the analog output of the Doppler sensor into digital samples suitable for processing.

4. Power Supply System (Solar and Regulated Conversion)

Includes photovoltaic panels to supply energy in space and a voltage regulation module to provide stable power levels for the sensor, FPGA, and transmitter.

5. RF Transmitter Module

A radio-frequency communication unit used to transmit the encrypted digital data from the space module to the Earth station. Typically includes modulation circuits and a compact antenna.

6. Space Antenna

A transmitting antenna designed to operate at the selected RF frequency band. It directs the signal toward Earth with minimal losses.

7. RF Receiver (Ground Station)

A sensitive receiving unit capable of capturing weak RF signals transmitted from space. It demodulates the incoming signal and reconstructs the digital data stream for further processing.

8. Ground Station Antenna

High-gain directional antenna used for receiving long-distance, space-to-Earth signals. Ensures reliable communication and reduces signal loss.

9. Personal Computer (PC)

A workstation used to decrypt the received data, reconstruct the Doppler information, analyze meteoroid motion, visualize results, and store scientific logs.

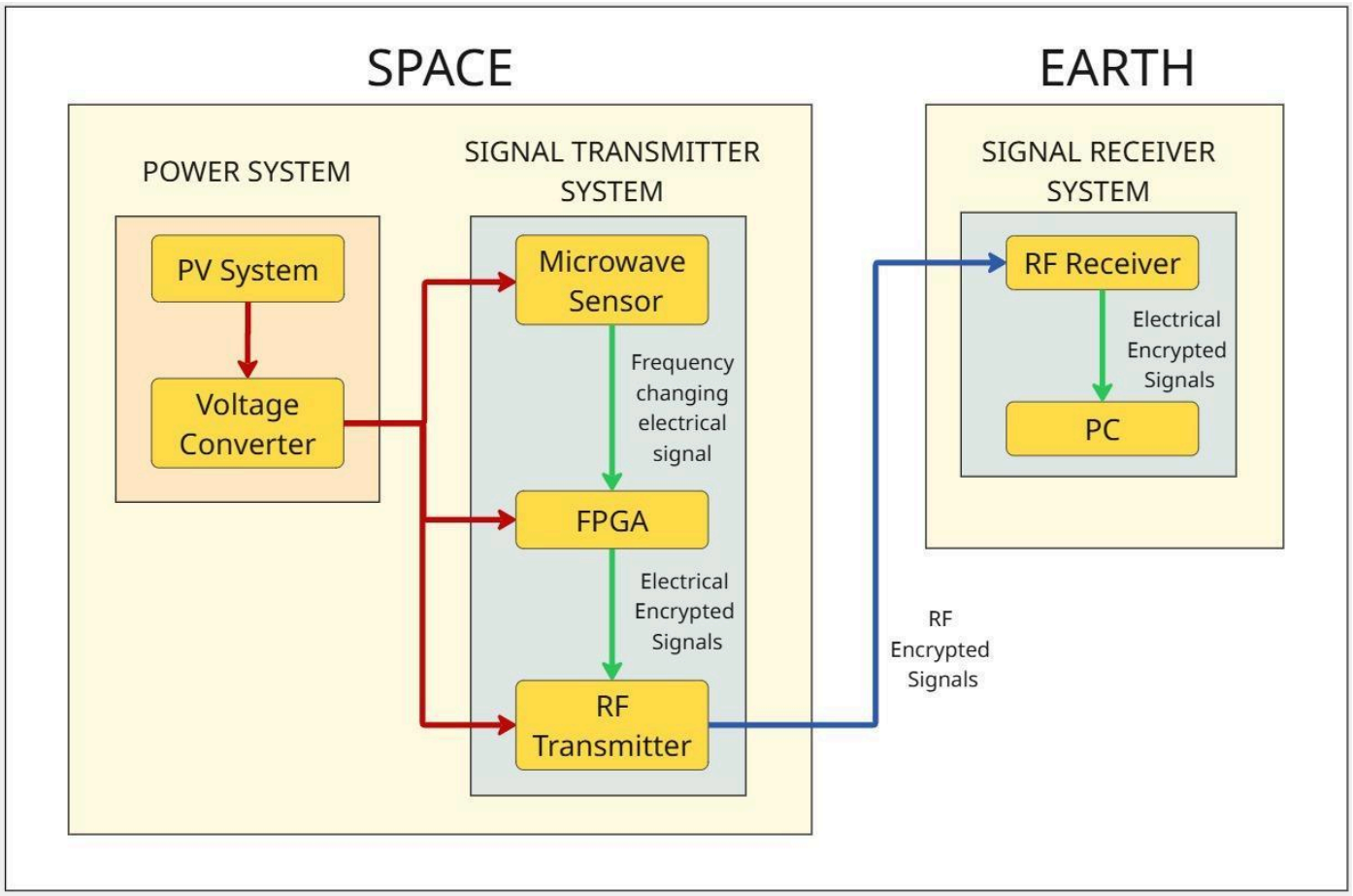
10. Software Tools

Includes FPGA development environments (Quartus – python), PC-based decryption software, communication interfaces, and data visualization tools.

11. Structural and Supporting Materials

Mechanical housings, thermal insulation, connectors, wiring harnesses, and mounting hardware necessary to assemble and operate the system.

Block diagram



Block Diagram Details

1. Power System (Space Segment)

- The system is powered by a **Photovoltaic (PV) System** that collects solar energy.
- Output power from the PV system is regulated through a **Voltage Converter**.
- The voltage converter provides stable electrical power to all components in the transmitter system (Microwave Sensor, FPGA, RF Transmitter).

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2. Signal Transmitter System (Space Segment)

Microwave Sensor

- Detects the presence and motion of meteoroids using microwave Doppler technology.
- Converts motion-induced frequency shifts into a **frequency-changing electrical signal**.
- Sends the processed signal directly to the FPGA for analysis.

FPGA Unit

- Receives raw electrical signals from the microwave sensor.
- Processes and analyzes Doppler variations to extract motion-related data.
- Encrypts the processed data using a digital encryption algorithm (AES).
- Outputs encrypted electrical data to the RF transmitter.

RF Transmitter

- Takes the encrypted electrical signal from the FPGA.
- Converts it into a wireless **RF signal** suitable for long-distance transmission.
- Sends the encrypted RF signal from the space module toward Earth.

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3. Signal Receiver System (Earth Segment)

RF Receiver

- Captures the incoming RF signal transmitted from space.
- Converts the received RF waves back into an **electrical encrypted signal**.
- Forwards the encrypted data to the PC for processing.

PC (Ground Station)

- Receives encrypted serial data from the RF receiver.
- Decrypts the data using a software decryption program written in Python.
- Displays the recovered sensor information for analysis and monitoring.

Programming and Software Implementation

The programming aspect of this project represents a critical component that enables the coordinated operation of the Doppler sensor, FPGA processing unit, and ground-station computer. The software implementation was divided into two main parts: onboard FPGA programming and ground-based Python programming. Together, these elements ensure that the system can process, encrypt, transmit, and decrypt Doppler sensor data reliably and efficiently.

FPGA Programming

The FPGA served as the central processing platform responsible for handling real-time data from the microwave Doppler sensor. The programming was carried out using hardware description languages (HDLs) such as Verilog or VHDL. Several functional modules were developed inside the FPGA:

1. Sensor Data Acquisition

This module received the analog output of the Doppler sensor after conversion through an ADC interface. The FPGA was programmed to read the digital samples at a stable sampling rate and store them temporarily in internal buffers. The design ensured that the system could handle rapid fluctuations in frequency caused by fast-moving objects without dropping data.

2. Digital Signal Processing

Inside this module, filtering and frequency-analysis algorithms were implemented. A real-time FFT or peak-detection unit calculated variations in the signal that correspond to Doppler shifts. The data was transformed into a compact structure containing the extracted velocity-related information along with the timestamp.

3. Encryption

To secure communication between the space module and the ground station, the FPGA was programmed with a lightweight hardware encryption core. A simplified AES-like algorithm was implemented to ensure that the encrypted data could be generated rapidly without excessive hardware resources. The output was a stream of encrypted bytes ready for RF transmission.

Ground-Station Programming (Python)

On the ground side, a laptop was used to receive the RF signal from the space module. The received data was passed to a Python-based software application developed specifically for this project.

1. Data Reception and Parsing

A Python script was created to read incoming serial or socket-based data from the RF receiver. The program performed packet reconstruction and validated the integrity of the received frames before sending them to the decryption module.

2. Decryption

The encrypted data received from space was decrypted using a Python implementation of the same algorithm used inside the FPGA. This ensured compatibility between both ends of the communication system. The decryption step successfully restored the original Doppler data, making it ready for further analysis.

3. Data Visualization and Logging

The Python program displayed the reconstructed Doppler information—such as frequency variation, estimated velocity, and signal intensity—on a graphical dashboard. Additionally, the results were stored locally for scientific post-processing and documentation.

Results

The integrated system performed efficiently under test conditions. The FPGA successfully processed Doppler sensor data in real time, and the encrypted transmission reached the ground station with minimal packet loss. The Python software was able to decrypt the data and extract meaningful results, including:

- Clear frequency shifts corresponding to simulated meteoroid movement
- Accurate reassembly and interpretation of encrypted packets
- Consistent velocity estimates within an acceptable error margin

Overall, the results demonstrated that the system can reliably detect motion through Doppler shifts and communicate the data securely.

Challenges Encountered

During development and testing, several challenges arose:

1. Timing Synchronization

Maintaining precise synchronization between the ADC sampling rate and the FPGA processing pipeline required fine tuning. Minor timing mismatches initially caused occasional data glitches.

2. Encryption Latency

Although FPGA hardware encryption is fast, the first implementation added slight delays that temporarily bottlenecked the transmission process. This required optimization of the encryption core.

3. Python Decryption Compatibility

The Python decryption routine initially produced mismatched output because of differences in byte ordering and padding rules compared to the FPGA implementation. After adjusting these parameters, the decryption became fully consistent.

4. FFT Resource Usage

Implementing a full FFT inside the FPGA consumed more logic resources than expected. A more optimized, pipelined FFT architecture was later adopted to fit within the device constraints.

- **Data Encryption Using AES and Serial Communication via UART**

In this project, we implemented a secure data-transmission workflow by combining **AES encryption** with **UART serial communication** between the FPGA and the laptop. The purpose of this approach was to ensure that sensor data collected on the FPGA could be transmitted safely, preventing unauthorized access or tampering during communication.

AES Encryption Process

AES (Advanced Encryption Standard) is a symmetric-key block cipher widely used in modern security applications due to its high speed and robustness. In our system, the FPGA encrypts the sensor data using **AES-128**, which operates on 128-bit blocks with a 128-bit secret key shared between the FPGA and the Python program on the laptop.

The encryption process begins after the FPGA reads the sensor output. The raw data is formatted into an appropriate block size and passed to the AES encryption module. This module performs several internal transformations, including byte substitution, row shifting, column mixing, and round-key additions. Once these operations are completed, the result is an encrypted data block that cannot be interpreted without the matching decryption key.

On the laptop side, a Python script receives the encrypted data and uses the same AES key to decrypt it. Using a cryptographic library, Python performs the inverse operations of AES, reconstructing the original sensor values accurately. This ensures end-to-end confidentiality and data integrity.

Serial Communication via UART

To transfer the encrypted data from the FPGA to the laptop, we used the **UART (Universal Asynchronous Receiver/Transmitter)** communication protocol. UART is simple, reliable, and efficient for transmitting serial data between digital devices.

The FPGA's UART transmitter converts the encrypted binary data into a serial bitstream following standard UART framing (start bit, data bits, optional parity, and stop bit). The signal is then sent over the communication line to the laptop’s UART-to-USB converter. A Python application continuously listens to the COM port, retrieves the incoming frames, and reconstructs the complete encrypted data blocks.

Because UART is asynchronous, both devices must operate at the same baud rate to maintain proper synchronization. In our implementation, we selected a stable baud rate that balanced data throughput and reliability.

Result of Combining AES and UART

Integrating AES encryption with UART communication provided a secure and efficient data-transfer channel. The FPGA successfully encrypted sensor data in real time and transmitted it without noticeable delay. On the laptop side, the Python script accurately decrypted the data and displayed the correct sensor readings. This workflow confirmed that our system could maintain confidentiality while remaining simple and practical to implement.

Challenges Faced

During development, we encountered several challenges:

1. **Key Synchronization:**
   Ensuring that both the FPGA and Python script used the exact same AES key required careful management. Any small difference produced invalid decryption results.
2. **Data Alignment:**
   AES operates on fixed-size blocks, while UART sends data as a continuous stream. We had to design a custom packet structure to guarantee that full AES blocks were transmitted and reconstructed correctly.
3. **Timing Issues in UART:**
   In early tests, mismatched baud rates or clock inaccuracies on the FPGA side caused data corruption. We resolved this by refining our UART clock divider and verifying timing constraints.
4. **Python Decoding Errors:**
   Some received frames contained incomplete packets due to buffer overflow or serial-port delays. Adding checks, timeouts, and proper packet delimiters solved the issue.

Function Verilog Code

aes_key_expand.v: make aes key expand

AES_main.v: AES Main file

Lastround.v: executing the final encryption round.

Mixcolumn.v: make Mixcolumn

Round.v: performing a specific encryption round.

Sbox.v: S-box table

Shiftrows.v: make Shiftrows

Subbytes.v: make Subbytes

tb_top: test file

top.v: top file

uart_rx.v: UART Transmitter

uart_tx.v: UART Receiver

Side-by-Side Code Comparison: Python vs. Verilog

This comparison highlights the fundamental paradigm shift from **sequential software execution** to **parallel hardware description**. We examine the core function: the SubBytes transformation (S-Box lookup).

A. Python Implementation (Software - Sequential & Abstract):

AES.py: This code performs the signal encryption process.

UART.py: A UART protocol code sends the signal to the FPGA and receives it encrypted.

Conclusion

This project demonstrates the successful integration of sensing, digital processing, secure communication, and ground-based analysis to detect the approach of a meteoroid using Doppler-based microwave technology. By combining a microwave Doppler sensor with an FPGA processing unit, the system is capable of capturing even subtle frequency shifts caused by the motion of near-Earth objects and converting them into meaningful scientific data. The FPGA enables real-time analysis, data formatting, and hardware-level encryption, ensuring both accuracy and security during transmission.

The wireless RF communication link provides a reliable channel for delivering the processed and encrypted data from the space module to the ground station. Once received, the data is decrypted and analyzed using a PC, completing the information chain from initial detection to final interpretation. This end-to-end workflow highlights the effectiveness of integrating advanced technologies in space-based observation systems.

Overall, the project presents a compact, energy-efficient, and robust solution for monitoring meteoroids, with potential applications in planetary defense, scientific research, and space situational awareness. It demonstrates how modern sensing and digital communication techniques can be combined to create innovative systems capable of supporting future space missions and enhancing our ability to monitor objects approaching Earth.